Correction du partiel

Q1) $fv(x) = \{x\}$

$fv(n) = \emptyset$

$fv(e_1 + e_2) = fv(e_1) \cup fv(e_2)$

$fv(e_1 / e_2) = fv(e_1) \cup fv(e_2)$

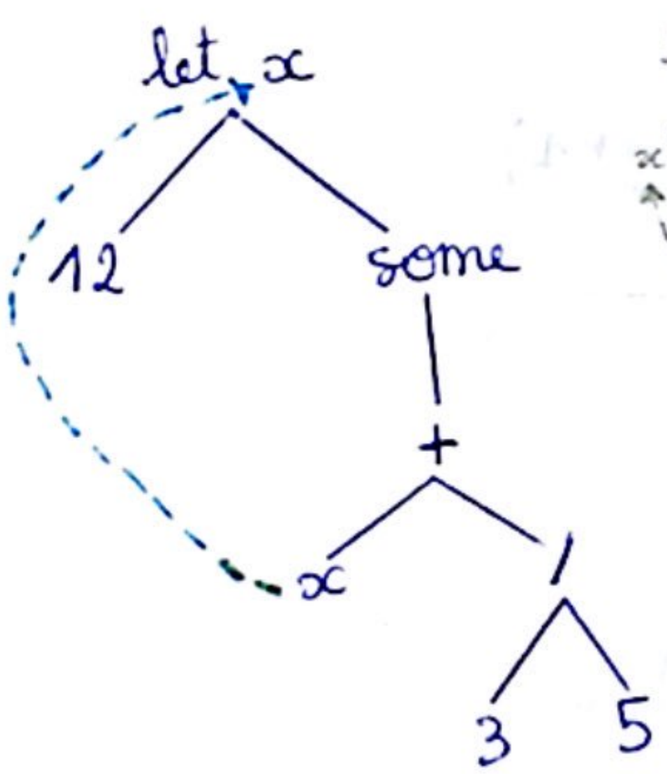
$fv(\text{let } x = e_1 \text{ in } e_2) = fv(e_1) \cup (fv(e_2) \setminus \{x\})$ ou $(fv(e_1) \cup fv(e_2)) \setminus \{x\}$

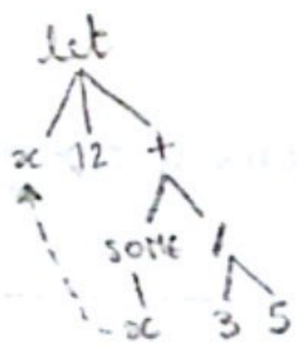
$fv(\text{None}) = \emptyset$

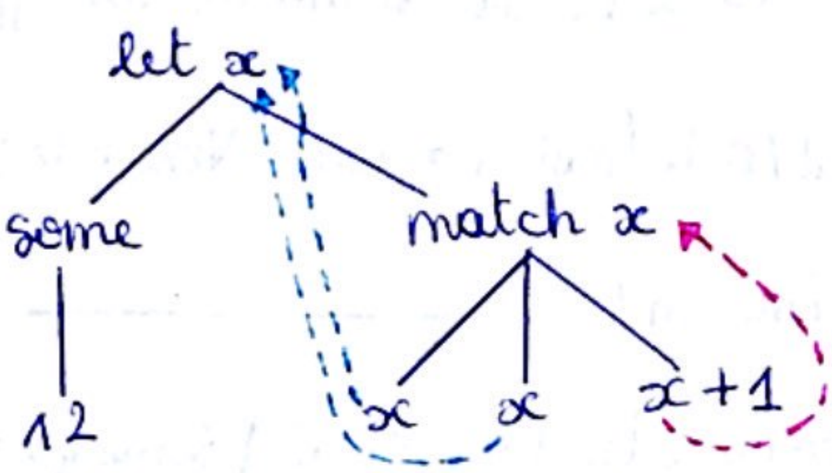
$fv(\text{some } e) = fv(e)$

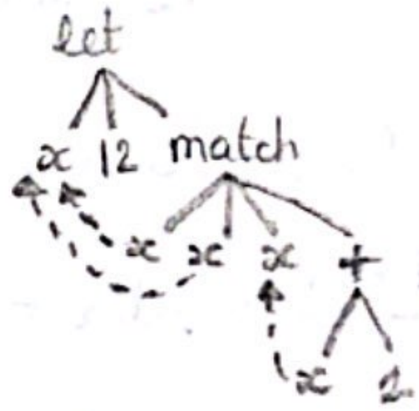
$fv(\text{match } e_1 \text{ with None} \Rightarrow e_2 \mid \text{some } x \Rightarrow e_3) = fv(e_1) \cup fv(e_2) \cup fv(e_3) \setminus \{x\}$

Q2)









Q3) • $(12 / 5) + 8$ Pas typable

DIV de type opt num

$$\text{ADD}\ \frac{\text{DIV}\ \dfrac{\Gamma \vdash 12 : \text{num} \quad \Gamma \vdash 5 : \text{num}}{\Gamma \vdash 12/5 : \text{num}} \quad \Gamma \vdash 8 : \text{num}}{\Gamma \vdash (12/5) + 8 :}$$

• some None Typable

$$\text{SOME}\ \frac{\text{NONE}\ \dfrac{}{\vdash \text{None} : \text{opt num}}}{\vdash \text{some None} : \text{opt (opt num)}}$$

• match 12/3 with None ⇒ None | Some x ⇒ x Pas typable

$$\text{MATCH}\ \frac{\text{DIV}\ \dfrac{\vdash 12 : \text{num} \quad \vdash 3 : \text{num}}{\vdash 12/3 : \text{opt num}} \quad \vdash \text{None} : \text{opt num} \quad x : \text{num} \vdash x : \text{num}}{\vdash \text{match } 12/3 \text{ with None} \Rightarrow \text{None} \mid \text{Some } x \Rightarrow x : \ ?}$$

opt num ou num

• let x = some 12 in (match x with None ⇒ x | some y ⇒ y/0) Typable

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$$\dfrac{\dfrac{\dfrac{}{\vdash 12 : \text{num}}\,\text{STR} \quad \dfrac{}{x : \text{opt num} \vdash x : \text{opt num}}\,\text{VAR}}{\vdash \text{some } 12 : \text{opt num} \quad x : \text{opt num} \vdash \text{match } x \text{ with None} \Rightarrow x \mid \text{Some } y \Rightarrow y/0}\,\text{SOME} \quad \dfrac{\dfrac{}{x : \text{opt num} \vdash x : \text{opt num}}\,\text{VAR} \quad \dfrac{\dfrac{}{x : \text{opt num}, y : \text{num} \vdash y : \text{num}}\,\text{VAR} \quad \dfrac{}{\vdash 0 : \text{num}}\,\text{STR}}{x : \text{opt num}, y : \text{num} \vdash y/0 : \text{opt num}}\,\text{DIV}}{\vdash \text{match } x \text{ with None} \Rightarrow x \mid \text{Some } y \Rightarrow y/0}\,\text{MATCH}}{\vdash \text{let } x = \text{Some } 12 \text{ in } (\text{match } x \text{ with None} \Rightarrow x \mid \text{Some } y \Rightarrow y/0) :}\,\text{LET}$$

Q4) On peut donner un contre-exemple.
Soit l'expression None qui est bien typé mais qui a plusieurs types.

$$\dfrac{}{\vdash \text{None} : \text{opt num}}\,\text{NONE} \qquad \dfrac{}{\vdash \text{None} : \text{opt (opt num)}}\,\text{NONE}$$

Ce qui valide l'absence d'unicité du typage.

Q5) • let $x = 1/0$ in (match x with None $\Rightarrow 0 \mid$ Some $y \Rightarrow y+1$)

LET + DIV0 ⟶ let x = None in (_____)

LETV ⟶ match None with None $\Rightarrow 0 \mid$ Some $y \Rightarrow y+1$

MATCHNONE ⟶ 0

• (match Some 12 with None $\Rightarrow 3 \mid$ Some $x \Rightarrow 2+x$) / 0

⟶ $(2+12)/0$ ⟶ $14/0$ ⟶ None

Q7) • let $x = ?\,4/2$ in let $y = ?\,6/3$ in x/y

⟶ let $x = ?$ Some 2 in let $y = ?\,6/3$ in x/y

⟶ let $y = ?\,6/3$ in $2/y$

⟶ let $y = ?$ Some 2 in $2/y$

⟶ $2/2$ ⟶ Some 1

• let $x = ?\,4/2$ in let $y = ?\,6/0$ in x/y

⟶ let $x = ?$ Some 2 in let $y = ?\,6/0$ in x/y

⟶ let $y = ?\,6/0$ in $2/y$

⟶ let $y = ?$ None in $2/y$ ⟶ None

Q8) $$\dfrac{\Gamma \vdash e_1 : \text{opt } \tau_1 \quad \Gamma, x : \tau_1 \vdash e_2 : \text{opt } \tau_2}{\Gamma \vdash \text{let } x = ?e_1 \text{ in } e_2 : \text{opt } \tau_2} \qquad \dfrac{e_1 \longrightarrow e_1'}{\text{let } x = ?e_1 \text{ in } e_2 \longrightarrow \text{let } x = ?e_1' \text{ in } e_2}$$

$$\dfrac{}{\text{let } x = ?\text{None in } e_2 \longrightarrow \text{None}} \qquad \dfrac{}{\text{let } x = ?\text{Some } v \text{ in } e_2 \longrightarrow e_2\{v/x\}}$$

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Q6) $e\{f/x\}\{v/y\} = e\{v/y\}\{f\{v/y\}/x\} \quad x \neq y$

x, n, None	cas de base
e_1+e_2, e_1/e_2, Some e, Match, let	cas d'induction

Si e est une variable

- $e = x$

$x\{f/x\}\{v/y\} = f\{v/y\}$

$x\{v/y\}\{f\{v/y\}/x\} = x\{f\{v/y\}/x\} = f\{v/y\}$

- $e = y$

$y\{f/x\}\{v/y\} = y\{v/y\} = v$

$y\{v/y\}\{f\{v/y\}/x\} = v\{f\{v/y\}/x\} = v$

- $e = z \quad z \neq x, \ z \neq y$

$z\{f/x\}\{v/y\} = z$

$z\{v/y\}\{f\{v/y\}/x\} = z$

Sans lieur

$-(e_1+e_2)\{f/x\}\{v/y\} \overset{\text{déf seulmt}}{=} e_1\{f/x\}\{v/y\} + e_2\{f/x\}\{v/y\}$

$= e_1\{v/y\}\{f\{v/y\}/x\} + e_2\{v/y\}\{f\{v/y\}/x\}$

$\overset{\text{déf seulmt}}{=} (e_1+e_2)\{v/y\}\{f\{v/y\}/x\}$

Hi x2

Avec lieur

$-(\text{let } z\ e_1 \text{ in } e_2)\{f/x\}\{v/y\} \overset{\text{déf seulmt}}{=} \text{let } z = e_1\{f/x\}\{v/y\} \text{ in } e_2\{f/x\}\{v/y\}$

$\overset{\text{HI x2}}{=} \text{let } z = e_1\{v/y\}\{f\{v/y\}/x\} \text{ in } e_2\{v/y\}\{f\{v/y\}/x\}$

$\overset{\text{déf seulmt}}{=} (\text{let } z = e_1 \text{ in } e_2)\{v/y\}\{f\{v/y\}/x\}$

$z \notin fv(f)$

$z \notin fv(v)$

$z \neq x, \ z \neq y$

$z \notin fv(f\{v/y\})$

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